AI-BASED ADAPTIVE VR MEDITATION SYSTEM

Comments	How we addressed them
VR Component How do you plan to evaluate your agent?	Designed a well-structured evaluation plan. Each user takes 3 sessions with 3 different conditions. Condition 1 – No CA-MAML, learns a global policy, Condition 2 – Only MAML, models pluralism, Condition 3 – Full CA-MAML pipeline, models pluralism + divergence Compare results for the 3 conditions with each other.
Stress classifier component Not confident enough about the component	Designed a well-structured pipeline. Trained and compared multiple models.
Audio controller component How do you plan to evaluate your agent?	Designed a well-structured evaluation plan similar to the VR component. 5 sessions with 5 conditions are tested instead of 3.
IoT component (Common component) Using PPG sensor to get the RR. Reducing the noise is the novelty	We clarified that the wearable device will use a MAX30102 PPG sensor to capture raw IR PPG signals. These signals will be processed into pulse peaks and IBI values, which are similar to RR intervals. Noise handling is addressed through signal cleaning, artifact removal, beat validation, and HRV feature extraction before stress prediction.

AI-BASED ADAPTIVE VR MEDITATION SYSTEM

Problem, It's Impact

- University students experience high stress due to academic workload, exams, deadlines, and career pressure.
- Stress affects concentration, sleep quality, emotional well-being, and academic performance.
- Many students do not actively seek help due to stigma, lack of time, or limited access to counselling services.

Who Will Benefit from the Solution

- Primary users: university students facing stress, anxiety, academic pressure, and mental fatigue.
- Universities: can provide a low-cost stress-relief tool through counselling units, wellness centers, or student support services.
- Counsellors and mental health support staff: can use the system as a supportive relaxation aid, not a replacement for therapy.
- Student communities: benefit from a more accessible, private, and engaging relaxation method.
- Technology benefit: mobile VR makes immersive relaxation more affordable than standalone VR headsets.

AI-BASED ADAPTIVE VR MEDITATION SYSTEM

Design Excellence of the Overall Solution

- Combines wearable-based stress detection, mobile app preferences, and adaptive VR relaxation into one integrated system.
- Uses a low-budget mobile VR approach, making the solution practical for university students.
- Personalizes the VR environment instead of giving every user the same fixed relaxation experience.
- Adapts the environment based on stress level and user preferences to improve comfort and relaxation.
- Provides a passive, easy-to-use VR experience suitable for students with no meditation experience.
- Supports future scalability through mobile app deployment, cloud backend, and university-level adoption.



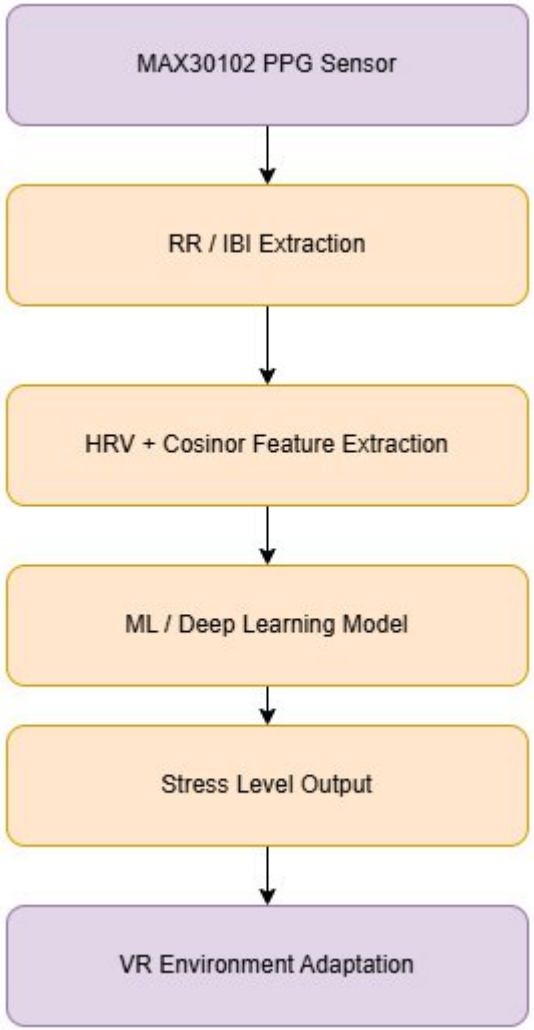
IT22191038 | PERERA B.D.P.D.

Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals

Specialization: Data Science

COSINOR-ENHANCED MULTI-LEVEL STRESS INFERENCE FROM WEARABLE RR INTERVALS

Prototype Workflow



Sub-Problem Addressed

This component focuses on identifying the user's stress level from physiological heart rhythm data.

How the Solution Addresses It

- Converts ECG/PPG-derived heartbeat intervals into HRV features.
- Uses Cosinor rhythm modelling to identify deviations from expected physiological rhythm.
- Predicts 4 stress levels: Relaxed, Mild, Moderate, High.
- Sends stress output to the adaptive VR meditation system.

User Requirements Addressed

- Real-time or near real-time stress monitoring.
- Non-invasive wearable-based sensing.
- Personalized stress-aware meditation experience.
- Simple stress output understandable by users.
- Support for adaptive VR relaxation content.

COSINOR-ENHANCED MULTI-LEVEL STRESS INFERENCE FROM WEARABLE RR INTERVALS

Design Excellence / Contribution

- Introduces a Cosinor-guided rhythm-aware stress inference approach.
- Improves stress prediction by comparing current HRV against expected rhythm baseline.
- Evaluates multiple models: XGBoost, CNN-LSTM, transfer learning, and attention-based models.
- Uses subject-independent validation through Leave-One-Subject-Out Cross-Validation.
- Supports future wearable deployment using PPG-derived IBI data.

Prototype Contribution

- Provides the intelligence layer for the VR meditation system.
- Enables the system to adapt meditation content based on predicted stress level.
- Bridges wearable sensing, machine learning, and immersive mental wellness.

User Feedback on Prototype

- Users found stress-level output easy to understand.
- Adaptive feedback was considered useful for guided meditation.
- Wearable-based input was preferred because it is non-invasive.
- Suggested improvements include better real-time response, clearer visual feedback, and more personalized recommendations.



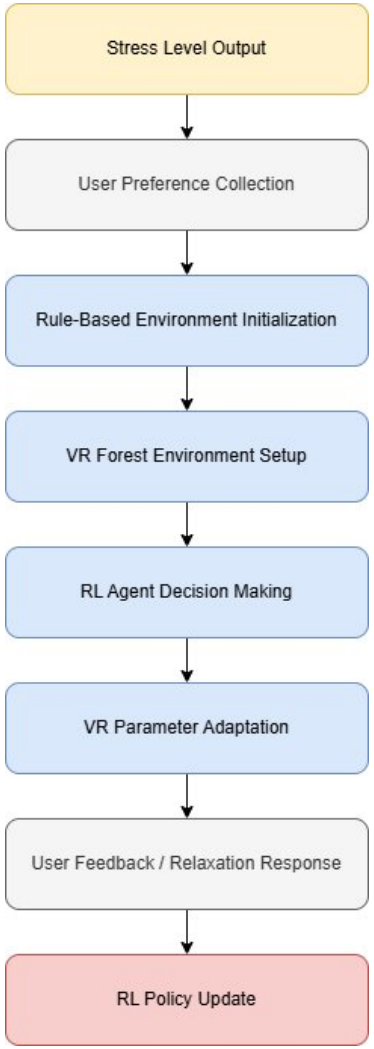
IT22231314 | PERERA W.T.S.R

Preference-Aware Adaptive VR Relaxation Engine

Specialization: Data Science

PREFERENCE-AWARE ADAPTIVE VR RELAXATION ENGINE

Prototype Workflow



How the Solution Addresses the Sub-Problem / Prototype

- This component focuses on adapting the VR relaxation environment based on user stress levels and preferences.
- User preferences collected through the mobile app are used to personalize the initial VR environment
- A rule-based mechanism sets the starting environment parameters before RL optimization begins
- The RL agent continuously adjusts controllable VR parameters to improve relaxation over time

User Requirements Addressed

- Personalized relaxation experience instead of a fixed VR scene
- Experience calming environments that match their preferences
- System should adapt when users respond differently to the same environment
- Solution should support passive VR relaxation without complex interaction
- Environment should remain comfortable, smooth, and non-distracting during the session

PREFERENCE-AWARE ADAPTIVE VR RELAXATION ENGINE

Design Excellence / Contribution

- Introduces adaptive personalization into a VR relaxation prototype
- Combines stress classification, user preferences, rule-based initialization, and RL-based optimization
- Models pluralistic and divergent user preferences, where different users may relax under different conditions
- Uses controllable VR parameters such as lighting, fog, weather intensity, sky color, and visual density
- Designed for a low-cost mobile VR setup using Unity and Android integration

User Feedback on Prototype

- Users liked the idea of the environment adapting according to their stress levels
- Had positive feedback on the overall aesthetic of the environment



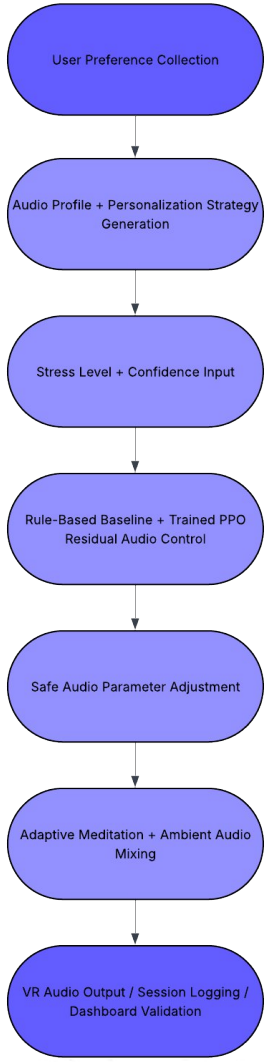
IT22172396 | SAMARASEKARA K.S.M.

CONFIDENCE-AWARE PERSONALIZED ADAPTIVE AUDIO CONTROL FOR AI-GENERATED VR MEDITATION

Specialization: Data Science

CONFIDENCE-AWARE PERSONALIZED ADAPTIVE AUDIO CONTROL FOR AI-GENERATED VR MEDITATION

Prototype Workflow



Sub-Problem Addressed

This component focuses on adapting meditation audio in real time to support relaxation using user audio preferences, stress level, and confidence-aware control.

How the Solution Addresses It

- Builds a personalized audio profile from user preferences
- Uses stress and confidence as runtime inputs
- Applies rule-based safe control plus trained PPO residual adaptation
- Adjusts intensity, density, brightness, and audio balance
- Blends meditation and ambient layers for immersive relaxation
- Logs actions, reward, and safety state for analysis

User Requirements Addressed

- Personalized meditation audio
- Stress-aware adaptation
- Safe low-confidence behavior
- Smooth and calming audio changes
- More immersive VR meditation
- Better personalization over time

CONFIDENCE-AWARE PERSONALIZED ADAPTIVE AUDIO CONTROL FOR AI-GENERATED VR MEDITATION

Design Excellence / Contribution

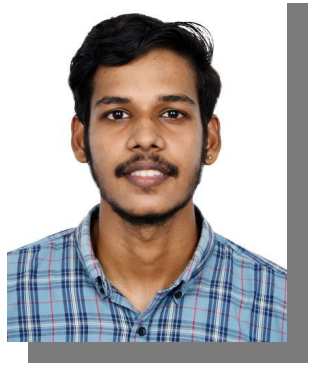
- Confidence-aware adaptive audio control for VR meditation
- Hybrid rule-based + trained PPO personalization
- Joint control of meditation and ambient audio
- Multi-parameter safe adaptation
- Novelty-aware variation to reduce habituation
- Validation dashboard for explainable adaptation

Prototype Contribution

- Provides the audio intelligence layer for the VR meditation system
- Connects user preferences, stress, confidence, and audio control
- Demonstrates safe and explainable adaptive playback in Unity
- Bridges offline RL training and Unity runtime playback
- Supports a more personalized relaxation experience

User Feedback on Prototype

- Compare adaptive audio vs static audio
- Measure smoothness and naturalness
- Check perceived personalization
- Assess comfort under confidence-aware adaptation
- Collect feedback on transitions and audio balance



IT22621788 | S. Prathikesh

VOICE-BASED COGNITIVE STRESS ASSESSMENT WITH CROSS-MODAL SESSION VALIDATION

Specialization: Data Science

VOICE-BASED COGNITIVE STRESS ASSESSMENT WITH CROSS-MODAL SESSION VALIDATION

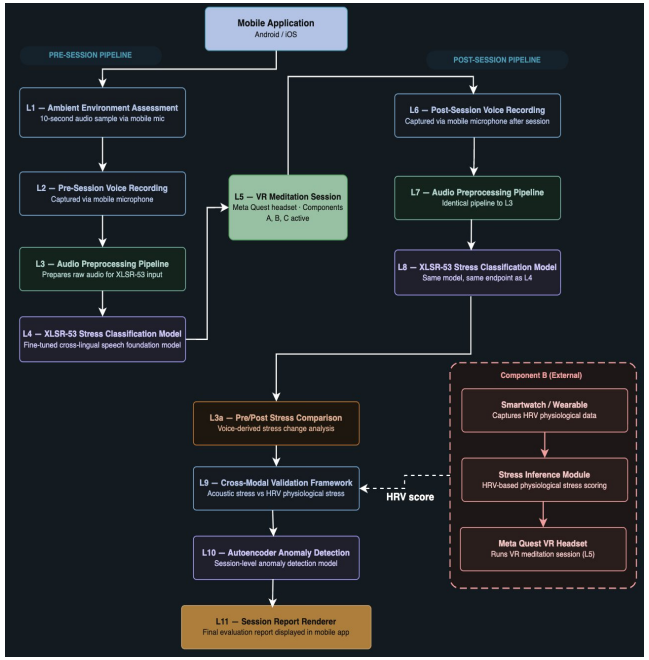
Sub-Problem

Traditional VR meditation systems guide users through relaxation sessions but do not evaluate whether the user's cognitive stress actually reduces after the session. A user may physically relax during VR meditation, but cognitive stress or mental pressure can still remain in their voice. Existing systems also do not combine voice-based stress assessment with HRV validation or session-level anomaly detection.

Proposed Solution

- Performs an ambient check to ensure reliable voice input
- Collects pre-session and post-session voice recordings
- Estimates cognitive stress using emotion2vec-based voice analysis
- Compares pre/post stress scores to identify stress change
- Validates the baseline result with HRV data from Component B
- Flags abnormal sessions using autoencoder-based anomaly detection
- Generates a student-friendly report with session history

Prototype Workflow



User Requirements Addressed by the Solution

- Simple guided voice recording process for university students
- Reliable stress assessment by blocking noisy recording environments
- Live microphone recording and uploaded audio file support
- Clear pre-session and post-session stress feedback
- Session effectiveness evaluation after VR meditation
- Research-based wellness feedback without claiming medical diagnosis

VOICE-BASED COGNITIVE STRESS & ENERGY ASSESSMENT WITH CROSS-MODAL SESSION VALIDATION

PROTOTYPE, USER FEEDBACK, EVALUATION AND ETHICS

Prototype / Solution Flow

- Cognify UI guides students through ambient check, pre-voice, VR session, post-voice and final report
- Both live microphone recording and uploaded audio are supported for each audio stage
- Final report shows stress change, HRV agreement and session flag without exposing raw JSON to normal users
- Voice stress inference uses emotion2vec embeddings with a classification layer

User Requirements Addressed

- Simple guided workflow for university students before and after VR meditation
- Clear environment suitability feedback before recording begins
- Student-friendly interpretation: not a medical diagnosis, only a research wellness estimate
- Saved session history supports repeated-session progress review

Evaluation and Validation Plan

- Evaluate stress classifier using accuracy, precision, recall and F1-score
- Use psychologist-guided or questionnaire-supported labels for student stress categories
- Validate pre/post stress reduction using score delta and chunk-level stress voting
- Report cross-modal agreement rate between voice baseline and HRV baseline
- Measure autoencoder anomaly detection using reconstruction error and threshold-based session flagging

User Feedback, Security and Ethics

- Prototype feedback will check clarity of instructions, recording comfort and usefulness of the final report
- Audio collection requires informed consent, anonymisation and minimal retention of raw voice data
- The system must clearly state that it is not a clinical diagnosis
- Psychologist guidance is required to define stress prompts, labels and ethical boundaries

AI-BASED ADAPTIVE VR MEDITATION SYSTEM

Customers

- University students experiencing academic stress and burnout

Market Size

- Growing demand for digital mental wellness and stress-management solutions
- VR relaxation solutions are becoming popular in wellness, healthcare, and education sectors
- Scalable market because the solution can be deployed through mobile VR with lower hardware cost

Project's Uniqueness

- Combines physiological stress detection with adaptive VR relaxation
- Personalizes the VR environment using user preferences and stress levels
- Personalizes the VR environment audio using user preferences and stress levels
- Uses AI/RL to adapt the environment instead of keeping it static
- Supports low-cost mobile VR instead of requiring expensive standalone headsets